

Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 1

What is Wearable Informatics?

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Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Definition

in•for•ma•t•ics

plural noun

1. the science of processing data for storage and retrieval;
informatics science

origin

- ▶ 1960s: from information + -ics, translating Russian **informatika**.

What is Wearable Informatics?

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Defined: Information

The Syllabus

Definition

1. facts provided or learned about something or someone: a vital piece of information.
 - ▶ Law a formal criminal charge lodged with a court or magistrate by a prosecutor without the aid of a grand jury: the tenant may lay an information against his landlord.
2. what is conveyed or represented by a particular arrangement or sequence of things: genetically transmitted information.
 - ▶ Computing data as processed, stored, or transmitted by a computer.
 - ▶ (in information theory) a mathematical quantity expressing the probability of occurrence of a particular sequence of symbols, impulses, etc., as contrasted with that of alternative sequences.

What is Wearable Informatics?

INF632
(EE499/EE599)

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Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Definition

- ▶ “denoting or relating to a computer or other electronic device that is small or light enough to be worn or carried on one’s body”
- ▶ “the science of processing data for storage and retrieval”
- ▶ Then: Wearable Informatics is the science of data storage, retrieval, and analysis from electronic or computer devices small enough to be worn”

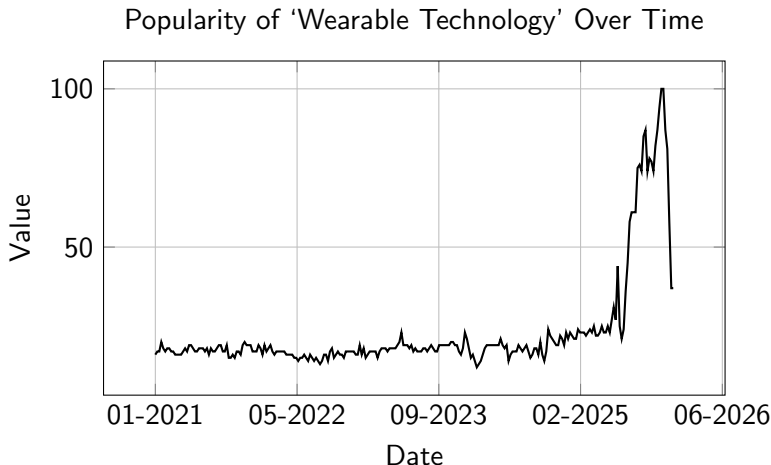


Figure: Time Series Data Visualization

Related Search Topics

- ▶ Technology (100)
- ▶ Watch (68)
- ▶ Health (63)
- ▶ Wearable computer (59)
- ▶ Sensor (52)
- ▶ Artificial intelligence (46)
- ▶ Data (39)
- ▶ Physical fitness (27)
- ▶ Smartwatch (24)
- ▶ Medicine (23)
- ▶ Sleep (23)

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Why?

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

► Use group data to:

► Gain scale

We can do analyses based on high resolution data sets, such as minute level observations (1440 minutes in a day), or even second level (86,400 seconds in a day)!

► Gain new insights

As researchers, and as end users, we have never before had access to so much data. Consider how much data we would have with 20 participants in a study that last 16 weeks (193M seconds worth)... What about all Fitbit users (large n), there are over 120 million of them!¹

► Make decisions

With this much data, we can make new comparisons between groups of people, especially when looking at health outcomes.

¹<https://electroiQ.com/stats/fitbit-statistics/>

Why? You!

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

- ▶ Use personalized data to:
 - ▶ Make decisions
- Imagine taking the findings of 120 million users and using that to personalize health care recommendations or guidance.

Current Headlines

Amazon On The Future For \$50 AI Wearable That Listens To Conversations

Forbes | 4 hours ago

You won't find it on Amazon's store just yet, but it sounds like the tech giant has big plans for subsidiary Bee's \$50 listening AI band.

Why Amazon bought Bee, an AI wearable | TechCrunch

TechCrunch | 2 hours ago

Amazon explains where its wearable Bee fits in and whether it will merge with Alexa.

Robotic puppy, AI, wearable technology on display at CES media preview

Las Vegas Review-Journal | 8 days ago

A robotic puppy is among the products thousands of CES attendees will see at multiple trade-show venues across Las Vegas starting Tuesday morning.

Can a Brain-Sensing Gaming Headset Finally Make Me a Pro Player?

gizmodo.com | 6 hours ago

I tried out Neurale and HyperX's new EEG gaming headset to become the best esports player I never wanted to be.



Japanese scientists' 'living skin' implant glows to warn of illness

The Messenger | 7 hours ago

The next generation of wearable health devices has been successfully trialed in mice by scientists in Japan.



AI pendants back in vogue at CES after early setback

RTE Online | 13 hours ago

Pendants and brooches packed with artificial intelligence abounded at CES 2026, using cameras and microphones to watch and listen through the day like a vigilant personal assistant.



Technology can be a great tool for boosting your health, but motivation is the foundation of sustained change

phillyvoice.com | 23 hours ago

Before buying a wearable or exercise app, take time to identify the inspiration and structure needed to build — and maintain — new habits in 2026.



Watch your fitness: six of best tech wearables to help you achieve your goals

Irish Daily Mirror | 1 day ago

From screen-free trackers to high-end adventure sport watches, we've got you covered for your 2026 health and fitness drive



The Neurotechnology Shift: how next-generation wearables interface with the brain itself

The Conversation en Español | 5 days ago



Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Undergraduate Course Objectives

Students who complete the undergraduate section of this course in good standing should be able to demonstrate the following advanced competencies:

1. Select, assess, and apply techniques appropriate to the application of wearable technologies for healthcare and wellness applications;
2. Synthesize, apply, and evaluate offline data analysis and machine learning techniques on large-scale data sets collected by wearable technologies;
3. Identify, interpret, and critically explain the significance of open research areas in wearable technologies and their applications in health-driven research;
4. Evaluate the applicability of wearable technologies in the commodity market.

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

Graduate Course Objectives

Students who complete the graduate section of this course in good standing should be able to demonstrate the following advanced competencies:

1. Select, assess, and apply techniques appropriate to the design and implementation of wearable technologies for healthcare and wellness applications, including specialized communication protocols and data structures and storage techniques;
2. Synthesize, apply, and evaluate online and offline data analysis and machine learning techniques on large-scale data sets collected by wearable technologies;
3. Identify, interpret, and critically explain the significance of open research areas in wearable technologies and their applications in health-driven research;
4. Evaluate the applicability of wearable technologies in the commodity and research.

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

Who am I?



Kyle N. Winfree, PhD
Associate Professor
Faculty of Informatics,
Computer Science, and Electrical
Engineering in the School of Informatics,
Computing, and Cyber Systems
Joined NAU in Fall of 2015
AD of Undergrad Programs 2019-2022
AD of Grad Programs 2022-2025

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Formal Education and Training

- ▶ NIH mHealth Training Institute Fellow
- ▶ Post-doc at Univ. of Delaware, PD intervention and assessment (SEnsole)
- ▶ Ph.D. at Univ. of Delaware in Biomechanics and Movement Science
 - 1.5 years focused on stroke rehabilitation robotics (ALEX)
 - 2.5 years dissertation on Parkinson's disease rehabilitation (PDSHoe)
- ▶ M.S.E. at Univ. of Pennsylvania in Robotics, thesis on haptics (iTorqU)
- ▶ US Geological Survey Astrogeology Branch, studied polar ice on Mars
- ▶ B.S. at **Northern Arizona University** in Physics

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate

Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

PhD / Post Doc Topic: PDShoe / Sensole

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate

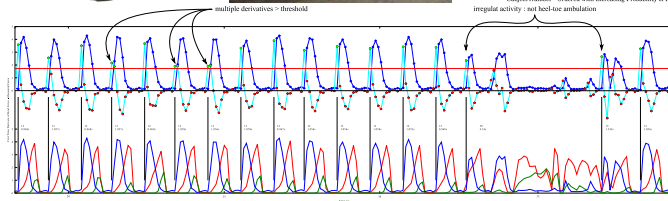
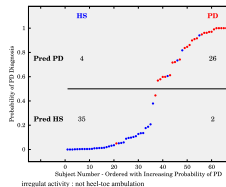
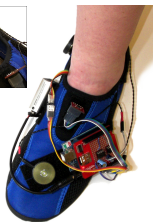
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?



MS Thesis : iTorqU

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate

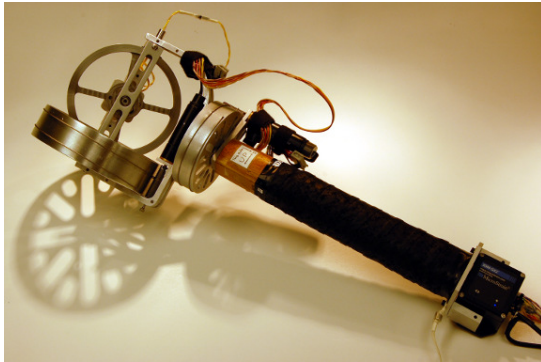
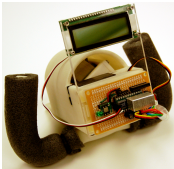
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?



Research interests

- ▶ Impact of Faculty Behaviors on Student-Faculty Rapport: A Multi-Institutional Study (2025)
- ▶ Optimizing Feature Extraction Methods Using Class Similarity Ratio for EMG-Based Hand Gesture Classification (2025, PhD Student)
- ▶ Ready, Set, Move! Tracking Children's Modified Ride-On Car Use With a Custom Data Logger (2023)
- ▶ Impact of Different Exercise Modalities on the Human Gut Microbiome (2021)
- ▶ Optimizing Student-Faculty Rapport for the Engineering Classrooms: Dimensioning the Behaviors that Matter (2020)
- ▶ The Development of an IoT Instrumented Bike: for Assessment of Road and Bike Trail Conditions (2018, PhD Student)
- ▶ Modeling Clinically Validated Physical Activity Assessments Using Commodity Hardware (2017)
- ▶ A novel method of assessing dietary behavior using a wrist-worn accelerometer (2017)

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate

Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

How do you contact me?

Email: kyle.winfree@nau.edu

Canvas: Better than email, send me a message on Canvas. That will pop up on my phone. I do NOT have email on my phone.

Office: SICCS Room 204

In-Person: Come see me in my office, stop me in the hallway, chat with me on a campus bus, etc. I prefer in-person discussions for anything more than a 2 minute question / response.

Cell Phone: 928.853.0114 (*please use this sparingly*)

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate

Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

Canvas and Git

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git

Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

- ▶ Due dates for assignments will be on Canvas
- ▶ Submission of assignments will be “linked” on Canvas (UG: Google Drive or GitHub, G: GitHub)
- ▶ Materials for class will be on GitHub/kylewinfree/inf632-spring2026
 - ▶ Source for all course materials will also be in Canvas. Graduate Students: I expect that you will typeset your assignment submissions; use this source to help you with $\text{\LaTeX}$.

Course Materials

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

- ▶ No text book will be required.
- ▶ You will need to purchase the materials to create a wearable device. This material list will be provided; expect costs to be approximately that of a textbook.
- ▶ Students in the undergraduate section have the option of completing the research project with off-the-shelf devices and may not need to purchase materials.

See the syllabus for recommended readings, including those that are available for free online.

Graduate and Undergraduate students will be assessed with different rubrics (expectations). The weight of each assignment is also different between these groups, owing to these different expectations.

Table: Assessments and related fractional percentage of final grade.

Assessment	Undergraduate	Graduate
Attendance and in-class participation	10%	10%
Homework Assignments (3)	45%	30%
Research Project: Literature Review	5%	10%
Research Project: Questions and Hypotheses	10%	10%
Research Project: Device Design and Implementation	5%	10%
Research Project: Methods Plan	5%	5%
Research Project: Analysis	5%	10%
Research Project: Discussion of Findings	10%	10%
Research Project: In-Class Presentation	5%	5%
Total	100%	100%

Wearable Informatics

Course Overview

The Syllabus

Grading

Class Composition and Goals

Participation Matters!

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading

Participation Matters!

Office Hours
Schedule

Class Composition and Goals

Who are you?

I assume you are here with the intention to learn something. With that assumption, then let's consider the following from J. E. Stice, "Using Kolb's Learning Cycle to Improve Student Learning," Engineering Education, pp. 291-296, February 1987.

Retention	What one ...	Class Activity
10%	reads	Readings
26%	hears	Lectures
30%	sees	Figures, Drawings, Slides
50%	sees and hears	Lectures with Visuals
70%	says	Asking questions, Discussing papers
90%	says while doing	Hands-on activities, Project demos

So come to class, for every lecture, every lab, and every project "office" hours!

Office Hours

INF632
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Class is on Tuesday and Thursday

It seems like Thursday afternoon, Friday morning, or Monday might be good office hours candidates.

So let's survey you!

Thursday 12:45-2:00	?
Thursday 2:30-3:45	?
Friday 9:00-10:15	?
Friday 10:00-11:15	?
Monday 1:00-2:15	?
Monday 2:00-3:15	?

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!

Office Hours

Schedule

Class Composition and Goals

Who are you?

Schedule

See the syllabus for a complete and readable version =>

Table: Tentative Course Schedule.

Week of	Content
January 12	Course expectations, setup (GitHub or Google Drive), and peer introductions (come prepared to introduce yourself with something you learned this past semester or summer). Start your assigned reading — this should become habit!
January 19	Dimensions of functionality Haptic interfaces <i>NAU is closed on Monday in observance of MLK Day</i>
January 26	Health research and trends in wearable health monitoring Analysis, comparative testing
February 2	Introduction of the Research Project Research methods for studies
February 9	Literature review, finding the gap in knowledge, forming a hypothesis No class on Thursday!
February 16	Statistical learning methods (applied) — regression, logistic regression, over fitting, neural nets
February 23	Statistical learning methods (applied) — support vector machines, decision trees, k-means, k-nearest neighbors
March 2	Device design Sensing
March 9	Spring Break! No class.
March 16	Soldering lab Arduino Part 1 Device design review (everyone shares)

INF632
(EE499/EE599)

Winfree

Wearable
Informatics

Defined: Wearable

Defined: Informatics

Defined: Information

Defined

Popularity

Why?

Current Headlines

Course Overview

Course Objectives

Undergraduate

Graduate

Instructor

The Syllabus

Canvas and Git

Course Materials

Grading

Participation Matters!

Classroom Etiquette

Schedule

Class Composition
and Goals

Who are you?

Who are you?

INF632
(EE499/EE599)

Winfree

Wearable Informatics

Defined: Wearable
Defined: Informatics
Defined: Information
Defined
Popularity
Why?
Current Headlines

Course Overview

Course Objectives
Undergraduate
Graduate
Instructor

The Syllabus

Canvas and Git
Course Materials
Grading
Participation Matters!
Office Hours
Schedule

Class Composition and Goals

Who are you?

1. Your name
2. Are you an Undergrad, Grad MS, Grad PhD, or... ?
3. Your Major
4. What might come after said degree?
5. Are you enrolled for credit or auditing?
6. Why are you here?
7. What are your research interests?
8. What is something you learned this past semester or summer?